# Hybrid method for well-mixed chemical reaction system

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# Abstract

A well-mixed chemical reaction system, described by the Master equation or equivalently, a stochastic equation involving Possion process, can be exactly simulated using the stochastic simulation algorithm. However, the algorithm become time-consuming as the frequency of reaction increase. In this paper, a deterministic approximation is derived from the Master equation, the error in the approximation is analyzed. Furthermore, a hybrid method coupling the stochastic method and the deterministic model is proposed. The hybrid method is applied to one simple chemical system.

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# Chapter 1

# Introduction

In this paper we concern with the following general problem: Consider a well-mixed mixture of $s_0$ chemical species, in a system of volumn N, which can inter-react through $r_0$ chemical reaction channels, then given the numbers of molecules of each chemical species at some initial time, what will these molecular population be at any later time?

A natrural way is to represent the system by the master equation [5], which is a system of coupled differential equations whose variable is the probabilities at some state, in systems with very few state, one may write down the entire system of equations and solve it, however, when the number of states is large, the Master equation becomes impossible to solve analytically and numerically.

In 1976, Gillesple [6] [7] proposed an simution algorithem, called stochastic simulation algorithm(SSA), which is fully equivalent to the Master equation, even though the Master equation is never explicitly used. At each iteration of the algorithm, the output is which reaction occurs next and when does it occur, so one can use SSA to simulate the chemical system up to any later time. However, when the frequency of reaction of the system is large, this algorithm is slow owing to the explicit time stepping. Gibson [1] speed up SSA by introducing some data structure and using random number repeatly, but this method can only go so far in terms of speeding up the simulations, and still be computationally intensive for realistic problems.

Kurtz[2] [4] rigorously proved that a Markov chain can be approximated by a deterministic equation. So one may assume that there are sufficiently many molecules that the number of molecules can be approximated as a continously varyling quantity that varies deterministically over time. In this method, one writes a coupled system of differential equations for the concentration of each species in terms of the concentrations of all others. By implementing some numerical methods one can solve ODE very efficiently. This approach assumes that the time evolution of the state of a

system is continous and deterministic. However, the system is obviously not continuous because any species will change with the form of jump when any reaction occur. The system is not deterministic either because we do not take the quantum effects and positions and velocities of each molecules into account. Thus, the deterministic approach works well only for such systems that the number of species is quite large so the fluctuations can be neglected.

In this paper, we propose a hybrid method which uses ODE dynamics to simulate fast reactions coupled with SSA dynamics to simulate reactions in which discreteness cannot be discounted. Moreover, our scheme does not explicitly keep track of fast and slow reactions, but rather using a indicator function to indicate which method will dominate the whole simulation. Namely, the hybrid method will perform ODE method in regions where all involved species in abundance, SSA method in regions where one of the involved species is in small concentrations, and a mixture of both in intermidiate regions. The indicator function, also called blending function is defined for each individual reaction and be chosen to take value 1 in regions of low concentration, 0 in regions where all involved species in this reaction is in abundance, and smoothly interpolates in between. By introducing such a blending function, our hybrid method no longer need periodic repartitioning. Recent work by Andrew Ducan [3] proposed a hyrbid method using SSA dynamics and Langevin dynamics, based on the same region partitioning.

# Chapter 2

# Mathematical Descriptions of Reaction Process

Consider a well-mixed mixture of $s_0$ chemical species,in a system of volumn N,the *network* of $r_0$ chemical reactions involving $s_0$ chemical species, $S_1, ..., S_{s_0}$ is:

$$\sum_{i=1}^{s_0} v_{ik} S_i \rightharpoonup \sum_{i=1}^{s_0} v'_{ik} S_i, \qquad k = 1, ..., r_0 \qquad v_{ik}, v'_{ik} \in \mathbb{Z}^+$$

$v_{ik}$ : the coefficients of $i$th spiecies in $k$th reaction before reaction
$v'_{ik}$ : the coefficients of $i$th spiecies in $k$th reaction after reaction
Assuming $P\{reaction\ occurs\ in\ (t, t+\triangle t]|F_t)\} = \lambda_k(X(t))\triangle t$, the state of the system at time t satisfies:

$$X(t) = X(0) + \sum_k R_k(t)(v'_k - v_k) = X(0) + \sum_k Y_k(\int_0^t \lambda_k^N(X(s))ds)(v'_k - v_k) \quad (2.1)$$

where $R_k(t)$ is the number of times that $k_{th}$reaction occurs before time t, $Y_k$ is unit-rate Possion process, $\lambda$ is the reaction rate of the reaction, by law of mass action, we have:

$$\lambda_k^N(x) = \alpha_k \frac{\prod_i v_{ik}!}{N^{|v_k|-1}} \prod_i \binom{x_i}{v_{ik}}$$

where $|v_k| = \sum_i v_{ik}$.

# Chapter 3

# Deterministic approximation

3.1 *ODE approximation*

Let $\mathbb{Z}^d$ be the d-dimensional integer lattice. $X_N(t)$ be a Markov chain with state space $(1/N)k : k \in \mathbb{Z}^d$, be the solution of the stochastic equation:

$$X_N(t) = X_N(0) + \sum_l \frac{1}{N} l Y_l(N \int_0^t f_l^N(X_N(s))ds) \tag{3.1}$$

We assume $X_N(t)$ is uniquely determined by the infinitesimal parameters.
Let X(t) be the solution of the ordinary differential equation:

$$X(t) = X(0) + \int_0^t \sum_l l f_l(X(s))ds \tag{3.2}$$

$$= X(0) + \int_0^t F(X(s))ds \tag{3.3}$$

*Theorem:* If

$$|F(x) - F(y)| \leq M|x - y|, \sum_l |l|\varepsilon_l < \infty \tag{3.4}$$

and $\lim_{N\to\infty} X_N(0) = X(0)$,then

$$\lim_{N\to\infty} \sup_{t\leq T} |X_N(t) - X(t)| = 0 \quad a.s. \tag{3.5}$$

for every $T > 0$.
*Proof:* Proved by Thomas G.Kurtz in [4].
Now by the general theorem above, we invoke the deterministic form of mass action kinetics:

Set $z = N^{-1}X(t)$, notice that $\lambda_k^N(x) \approx N\alpha_k \prod_i z_i^{v_{ik}}$, we define $\hat{\lambda}_k(z) = \alpha_k \prod_i z_i^{v_{ik}}$, then we have:

$$z(t) = z(0) + \sum_k N^{-1}Y_k(\int_0^t \lambda_k^N(X(s))ds)(v'(k) - v(k)) \tag{3.6}$$

$$\approx z(0) + \sum_k N^{-1}Y_k(N\int_0^t \hat{\lambda}_k(z(s))ds)(v'(k) - v(k)) \tag{3.7}$$

$$= z(0) + \sum_k N^{-1}\hat{Y}_k(N\int_0^t \hat{\lambda}_k(z(s))ds)(v'(k) - v(k)) + \int_0^t F(z(s))ds \tag{3.8}$$

Where $\hat{Y}_k(u) := Y_k(u) - u, F(z) := \sum_k \alpha_k \prod_i z_i^{v_{ik}}(v'_k - v_k)$
Set $u = \int_0^t \hat{\lambda}_k(z(s))$, by the law of large numbers for the Possion process,

$$N^{-1}\hat{Y}(Nu) = N^{-1}(Y_k(Nu) - Nu) \tag{3.9}$$

$$= \frac{Y_k(Nu)}{N} - u \to 0, N \to \infty \tag{3.10}$$

Then (3.3) becomes

$$z(t) \approx z(0) + \sum_k \int_0^t F(z(s))ds \tag{3.11}$$

$$= z(0) + \sum_k \int_0^t \alpha_k \prod_i z_i(s)^{v_{ik}}(v'_k - v_k) \tag{3.12}$$

Since F(z) is continous,by Fundamental theorem of calculus,we get the coresponding ODE for the system:

$$z'(t) = \sum_k \alpha_k \prod_i z_i(t)^{v_{ik}}(v'_k - v_k) \tag{3.13}$$

Let $X_N(t) = \frac{X(t)}{N}$be the actual state of the system at time t

3.2 *Error estimate*

Follow the result of Kurtz [4] we have the error of such an diterministic approximation is:

$$X_N(t) = X(t) + O(\frac{1}{\sqrt{N}}), \tag{3.14}$$

where $X(t)$ be the solution of ODE(3.3).
This error estimate implies that in a given volumn system, relative error is small only when the number of species is abundant.

# Chapter 4

# The Hybrid scheme

4.1 *Blending functions*

The blending function $\beta_1(x), \beta_2(x), ..., \beta_{r_0}(x)$, defined for every reation in the system, is a function of the number of species take value in [0,1]. Intuitively, blending function is used for taking weighted average of ODE mehod and SSA method, hopefully we can get a better trade-off between cost of computation time of SSA and computation error of ODE than both pure ODE and pure SSA. For system that involve only 1 species:

$$\beta(x, I_1, I_2) = \begin{cases} 1, & x \leq I_1 \\ \frac{I_2 - x}{I_2 - I_1}, & I_1 < x < I_2 \\ 0, & x \geq I_2 \end{cases} \tag{4.1}$$

where $0 < I_1 < I_2$ are the boundaries between the different regions. The hybrid process will perform purely SSA for for $0 < x(t) < I_1$, purely ODE when $I_2 < x(t) < \infty$, and the weighted average in between.
For reacions that involve multiple species, let $S_r$ be the set of species involved in r-$th$ reaction(Both as reactants and products), we define:

$$\beta_r(X) = 1 - \prod_{n \in S_r} (1 - \beta(X_n, I_1^n, I_2^n))$$

where $I_1^n < I_2^n, n = 1, 2, ..., s_0$, are the boundaries for each individual chemical species. Given a choice of blending function, for each reaction, if at any time there is one species involved in this reaction whose population is lower than $I_1$, the hybrid process will perform pure SSA in this reaction, if all species involved in this reaction is in abundance($> I_2$), the hybrid process will perform pure ODE in this reaction, otherwise the hybrid process will blend beween ODE and SSA to simulate the corresponding reaction. The choice of boundaries $I_1^n, I_2^n$ is important to properly partition

the discrete region and continous region, which can be chosen empirically.

4.2 *The hybrid scheme*

We consider the following equation:

$$X(t) = X(0) + \sum_k Y_k(\int_0^t \beta_r(X(s))\lambda_k([X(s)])ds)(v'_k - v_k) \tag{4.2}$$

$$+ \sum_k (v'_k - v_k) \int_0^t (1 - \beta_r(X(s))\alpha_k \prod_i X_i(s)^{v_{ik}} ds, \tag{4.3}$$

where $N$ is the volume of the system, $[X(s)]$ is the closet point in lattice $\mathbb{Z}^{s_0}$ to X(s)$\in$ $\mathbb{R}^{s_0}$

*Remark* :In regions where $\beta_r(X) = 1$ the dynamics of the r-th reaction is modelled by the pure SSA, when $\beta_r(X) = 0$, the dynamics is pure ODE, in intermediate regions where $0 < \beta_r(X) < 1$ we abtain a convex combination of the two.

# Chapter 5

# Simulation of hybrid model

5.1 *Algorthm*

The input of the algorithm:

Propensities $\alpha_1, ..., \alpha_{r_0}$; blending functions $\beta_1, ..., \beta_{r_0}$; stoichimetric matrix for reactants: $v = (v_1, v_2, ..., v_{r_0})$, for products: $v' = (v'_1, v'_2, ..., v'_{r_0})$, final time of simulation T; time step for ODE:$\Delta t, \delta t(\delta t > \Delta t)$; initial state $X(0) \in \mathbb{Z}^{s_0}$

```
Set t = 0;
while t < T do
    if max_j β_j(X(t)) = 0 then
        Simulate ODE (6) with propensity α_i by Euler method up to time t + δt;
        Set t = t + δt;
    else if min_j β_j(X(t)) = 1 then
        Compute λ_0 = Σ_{j=1}^{r_0} λ_j([X(t)]);
        Generate r_1, r_2 from URN Take τ = (1/λ_0) ln(1/r_1);
        Take μ so that Σ_{v=1}^{μ-1} λ_v < r_2 λ_0 < Σ_{v=1}^{μ} λ_v;
        Set X(t + τ) = X(t) + v'_μ − v_μ;
        Set t = t + τ ;
    else
        Compute λ'_0 = Σ_{j=1}^{r_0} β_j(X(t)) λ_j([X(t)]);
        Generate r_1 from URN;
        Take τ = (1/λ'_0) ln(1/r_1);
        if τ < Δ t then
            Generate r_2 from URN;
            Take μ so that Σ_{v=1}^{μ-1} λ_v < r_2 λ_0 < Σ_{v=1}^{μ} λ_v;
            Simulate ODE (6) with propensity (1 − β_j(X)) α_j(X) up to time t + τ and set
              X(t + τ) = X(t + τ) + v'_μ − v_μ;
            Set t = t + Δt;
        else
            Simulate ODE (6) with propensity (1 − β_j(X)) α_j(X) up to time t + Δt;
            Set t = t + Δt;
        end
    end
end
```

**Algorithm 1:** Simulation of hybrid model

Equation (4.2) is a convex conmbination of stochastic equation (2.1) and core-

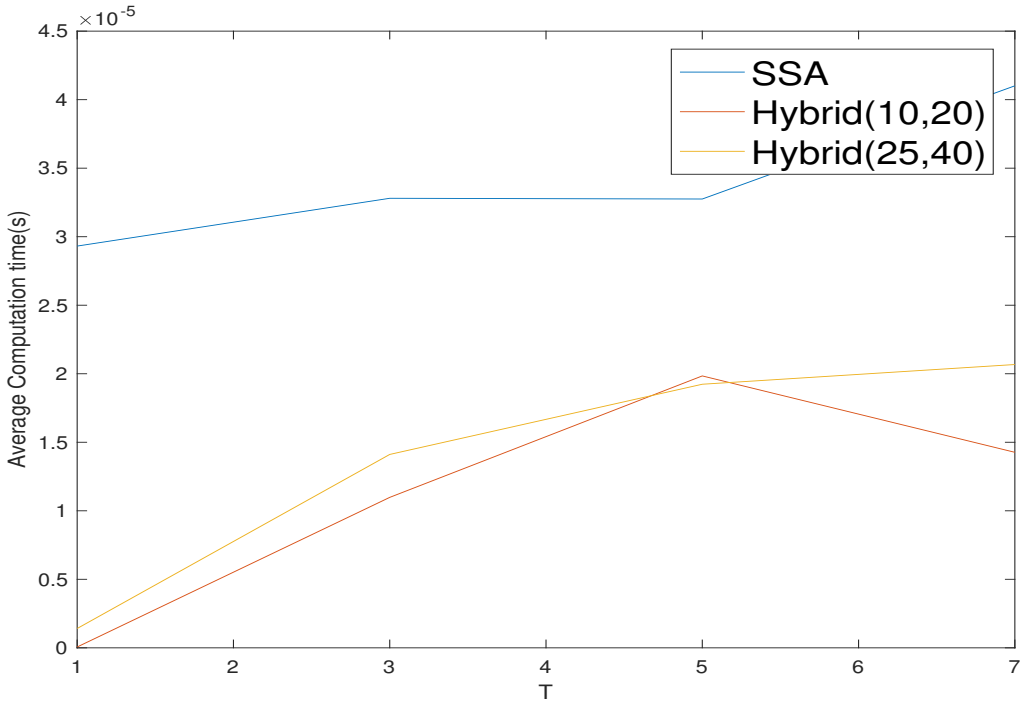


Figure 5.1: Average computation time as a function of T,X(0) = 60, h = 0.1

sponding ODE(3.8), which can capture both the discrete and continuum nature of the chemical system. We use SSA to simualte stochastic part and Euler method to simulate ODE part, this numerical scheme can generate realisations of SSA-ODE process.

5.2 *Numerical results*

The hybrid method is applied to the simulation of the simplest system with 1 speices:

$$A \rightharpoonup Z,$$

where Z is not a species of interest.

In Figure 5.1, we compare the average computational(CPU) time to simulate each model up to time T, each model averaged over 100 realisations. The computational cost of hybrid method was ploted for 2 different choice of boundaries of blending functions: $(I_1, I_2) = (10, 20), (25, 40)$, as T increases, the computation cost of the SSA scheme increases faster than the hybrid method.

In Table 5.1, we compare the average accuracy of SSA and ODE method, simulate to time T, we get the result of Hybrid method is more close to SSA than ODE, so Hybrid method is more accurate than ODE method.

| T | SSA | ODE | Hybrid(10,20) |
|---|---|---|---|
| 3s | 12.000 | 12.800 | 11.450 |
| 5s | 3.000 | 4.616 | 2.870 |
| 7s | 0.800 | 1.655 | 1.201 |

Table 5.1: Average simulation result of three methods with X(0) = 60,h=0.1

# Chapter 6

# Conclusion

In this paper, we propose a hybrid method for simulating chemical reaction system both efficient and accurate in some sense. Future work is needed to precisely quantify the trade-off between efficiency and accuracy, and to find the optimal transition.

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# Bibliography


[1] Michael A.Gibson and Jehoshua Bruck. Effcient exact stochastic simulation of chemical systems with many species and many channels. *J.Phys.Chem*, 104(2000):1876–1889, 1999.

[2] David F. Anderson and Thomas G. Kurtz. *Continuous Time Markov Chain Models for Chemical Reaction Networks*, pages 3–42. Springer New York, New York, NY, 2011.

[3] Andrew Duncan, Radek Erban, and Konstantinos Zygalakis. Hybrid framework for the simulation of stochastic chemical kinetics. *J. Comput. Phys.*, 326:398–419, 2016.

[4] Thomas G.KURTZ. Strong approximation theorems for density dependent markov chains. *Stochastic Processes and their Applications*, 6:223–240, 1978.

[5] G.W.Gardiner. Handbook of stochastic processes for physics,chemistry and natural sciences, 2nd edition. pages 100–122, 1985.

[6] Danlel T.Gillespie. A general method for numerically sim- ulating the stochastic time evolution of coupled chemical reactions. *J. Computational Phys.*, 22(4):403–434, 1976.

[7] Danlel T.Gillespie. Exact stochastic simulation of coupled chemical reactions. *J. Phys. Chem.*, 81(23):40–61, 1977.